

# Thirulok Sundar Mohan Rasu

Ann Arbor, MI 48105— +1 (248) 704-2911—[thirulok@umich.edu](mailto:thirulok@umich.edu)—  
[thiruloksundar.github.io](https://thiruloksundar.github.io)—[github.com/Thiruloksundar](https://github.com/Thiruloksundar)—[linkedin.com/in/thirulok-sundar-mohanrasu](https://linkedin.com/in/thirulok-sundar-mohanrasu)

## Education

### University of Michigan — Ann Arbor

Aug 2025 – May 2027

M.S., Electrical and Computer Engineering.

GPA: 4.0 / 4.0

Relevant courses: *Introduction to Algorithmic Robotics, Probability and Random Processes*

### Indian Institute of Technology (ISM) Dhanbad

Dec 2021 – May 2025

B.Tech., Electrical Engineering. Member: *CyberLabs (Machine Learning)*

GPA: 8.30 / 10

## Skills

**Languages:** Python (Advanced), C++, MATLAB.

**Areas:** ML, AI, Computer Vision, 3D/4D Reconstruction, Foundational models, LLMs, Motion planning, State estimation and Localization, Robotics Algorithms, Lidar, SLAM, Sensor Fusion, Probability, Statistics, Signal Processing

**Tools & Frameworks:** PyTorch, TensorFlow, Gazebo, RViz, AWS, Linux, Github, OpenCV, Simulink, PyBullet, ROS2, Nav2

## Experience

### University of Michigan, Ann Arbor

Aug 2025 – Present

*Research Assistant (Prof. Jeong Joon Park)*

- Creating a 4D dataset for robot perception and learning; designing data collection pipelines and evaluation protocols.

### Carnegie Mellon University

Jan 2024 – Sep 2025

*Research Intern (Dr. Arun Balajee Vasudevan)*

- Developed a benchmark audio dataset using multiple TTS models with paired real/fake audios; engineered pipelines for generation, labeling, and evaluation. **Paper (click here) published to ICASSP 2026 Conference.**
- Trained an audio–language detection model and used transcript-based detection (Llama-8b) to identify synthetic audio; team belief: our model is the best so far at identifying fake audios.
- Visualized audio waveforms and experimented with background noise removal/ speech enhancement techniques.

### University of Central Florida

Feb 2024 – Dec 2024

*Research Intern (Prof. Yogesh Rawat)*

- Mitigated activity hallucinations in Video-LLMs; created an annotated video-activity dataset and fine-grained evaluation metrics for actor/action/object/environment.
- Designed targeted question types and evaluation methods for robust activity understanding.

### Mowito

Mar 2023 – Jul 2023

*Robotics Software Intern*

- Implemented perceptual-hashing to deduplicate images in time-sorted training datasets, improving model training efficiency.
- Built a Dataset class supporting multiple annotation types (instance, picking, crate) and visualization tools; integrated into production pipelines.
- Implemented order-completion time estimation and Leiden clustering (repo link) for crate grouping to optimize order fulfillment.

## Projects

### Robot Localization With Kalman and Particle filters (GitHub link)

Implemented Kalman and Particle filters on PR2 robot under five different scenarios with interesting paths and studied their advantages and limitations

### Autonomous Mobile Robot Navigation System (GitHub link)

Developed complete autonomous navigation system for differential-drive mobile robot using ROS2 and Nav2 navigation stack; real-time obstacle detection and collision avoidance in C++ processing lidar data, enabling real-time collision avoidance

### Challenges in Multi-view 3D Scene Reconstruction (Paper link)

Analyzed failure cases of MAST3R for multi-view reconstruction under challenging capture conditions; collected a small scene dataset and documented limitations.

### Bar-Code Detection using OpenCV (GitHub link)

Classic CV pipeline to detect barcodes on grocery items and classify bounding boxes by barcode readability status.

## Selected Achievements

- Top-25 team, Smart India Hackathon 2024 (disaster-notification webpage with NLP).
- 54th All-India, Amazon ML Challenge 2023 (FAISS-based product dimension prediction) (Repo link).